Vectorizing Linear Time-Varying Recurrences

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Many image processing operations reduce to evaluating a **linear time-varying (LTV) recurrence**. [YZ: Insert definition LTV recurrence] For example, below is a second-order LTV recurrence:

$$f(x) = a(x)\, f(x-1) + b(x)\, f(x-2) + g(x)$$

where a, b, and g are coefficient arrays (provided or computed on-the-fly) that vary with position (time). A linear time-varying recurrences is the most general form of linear recurrences, and many IIR filters are instances of it. To compute it naively is straightforward, we can just write a simple for loop ranging over x, computing one pixel at a time. But to maximize single-core throughput on modern CPUs, we need to process multiple pixels simultaneously, with SIMD instruction sets like AVX or AVX-512. On the other hand, recurrences like this have a sequential dependency chain on the previous values. You cannot start computing $f(x)$ until you have $f(x-1)$ and $f(x-2)$. This forces a serial, one-pixel-at-a-time implementation.

There has been previously known algorithms to parallelize the computation of LTV recurrences, but it's not very practical. This post describes a new algorithm that breaks the dependency chain and vectorizes LTV recurrences with better practical constants than existing approaches.

The Classical Approach: Matrix Prefix Scans

The standard technique for parallelizing recurrences goes back at least to Karp, Miller, and Winograd (1967), later popularized by Blelloch as a general parallel primitive. The idea is to lift the scalar recurrence into matrix form. Observe that the update can be written as:

$$\begin{pmatrix} f(x) \\ f(x-1) \\ 1 \end{pmatrix} = \underbrace{\begin{pmatrix} a(x) & b(x) & g(x) \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{M(x)} \begin{pmatrix} f(x-1) \\ f(x-2) \\ 1 \end{pmatrix}$$

So to say compute 16 output pixels $f(x-15), f(x-14), \ldots, f(x)$ in parallel, we can compute the prefix product scan of $M(x-15), M(x-14), \ldots, M(x)$ with initial value $\begin{pmatrix} f(x-16) \\ f(x-17) \\ 1 \end{pmatrix}$. Because matrix multiplication is associative, its prefix scans can be easily parallelized: We can first compute $M_1(x) = M(x) \times M(x-1)$, then $M_2(x) = M_1(x) \times M(x-2)$, and so on, doubling the window size at each step until we have the full prefix product. Finally,

$$f(x) = M_{16}(x)_{1,1} * f(x-16) + M_{16}(x)_{1,2} * f(x-17) + M_{16}(x)_{1,3}$$

One observation is that the last row of every $M(x)$ is always $(0, 0, 1)$, and this is preserved under matrix multiplication. So rather than tracking all 9 entries of a 3×3 matrix through the prefix scan, we only need to track the first two rows, which is 6 values per position. Each combine step costs 2 multiplications per tracked value, for a total of 12 multiplications per step, and there are $1 + \log_2 k$ steps to vectorize k lanes.

The work multiplier compared to scalar evaluation is about[1] $\mathbf{6} \cdot (\mathbf{1} + \log \mathbf{k})$ per output element. For $k = 16$ (AVX-512), this works out to about $6 \cdot 5 = 30$ times the scalar work. This overhead makes it likely impractical for exposing parallel work.

A Better Decomposition

The bottleneck in the matrix scan is that each combine step needs to track 6 values (the two nontrivial rows of the matrix). What if we could reduce that to 4?

The key insight is that the matrix $M_k(x)$ has more structure than just its bottom row. It actually computes some redundant information in the matrix that we can get rid of.

We maintain two families of precomputed arrays across a logarithmic tower of layers, where each layer covers a window of twice the size of the previous one (similar to the M_k matrix). And we claim that, at each layer, exactly **4 arrays** suffice: two **propagation weights** d_k and d_{k-1}, and two **windowed partial sums** f_k and f_{k-1}. Let me explain each family.

The Propagation Weights d_k

Define $d_k(x)$ as the total weighted contribution of $f(x-k)$ to $f(x)$, summing over all valid **paths** from position $x-k$ to position x. A path is a sequence of steps that can either advance by 1 (incurring the local weight a) or advance by 2 (incurring the local weight b). Concretely:

[1] This is approximate because for $M_1 6$ we only need to compute the first row and the second row of M_1 is just $[a(x), b(x), g(x)]$.

- A 1-step from $y-1$ to y contributes a factor of $a(y)$.
- A 2-step from $y-2$ to y contributes a factor of $b(y)$.

For instance, $d_1(x) = a(x)$ (the only path from $x-1$ to x is a single 1-step), and $d_2(x) = b(x) + a(x)\,a(x-1)$ (either a direct 2-step, or two 1-steps). The value $d_k(x)$ is a kind of transfer function — it tells you how much $f(x-k)$ propagates into $f(x)$ through the recurrence, summed over all routing choices.

Building d_k inductively for even k

The construction is recursive. To compute $d_{2k}(x)$ (paths from $x-2k$ to x), we split at the midpoint $x-k$ and consider two mutually exclusive cases:

1. **Paths that pass through** $x-k$: These contribute $d_k(x-k) \cdot d_k(x)$ — first travel from $x-2k$ to $x-k$ (weight $d_k(x-k)$), then from $x-k$ to x (weight $d_k(x)$).
2. **Paths that 2-step over** $x-k$: These land at $x-k+1$ after arriving at $x-k-1$, skipping the midpoint. They contribute $d_{k-1}(x-k-1) \cdot b(x-k+1) \cdot d_{k-1}(x)$ — travel from $x-2k$ to $x-k-1$, then 2-step to $x-k+1$ (weight $b(x-k+1)$), then continue to x.

These two sets of paths are genuinely disjoint: any path either visits $x-k$ or skips over it. So we get the exact formula:

$$d_{2k}(x) = d_k(x) \cdot d_k(x-k) + d_{k-1}(x) \cdot b(x-k+1) \cdot d_{k-1}(x-k-1)$$

For example, at window size 16 (i.e., $k = 8$):

$$d_{16}(x) = d_8(x) \cdot d_8(x-8) + d_7(x) \cdot b(x-7) \cdot d_7(x-9)$$

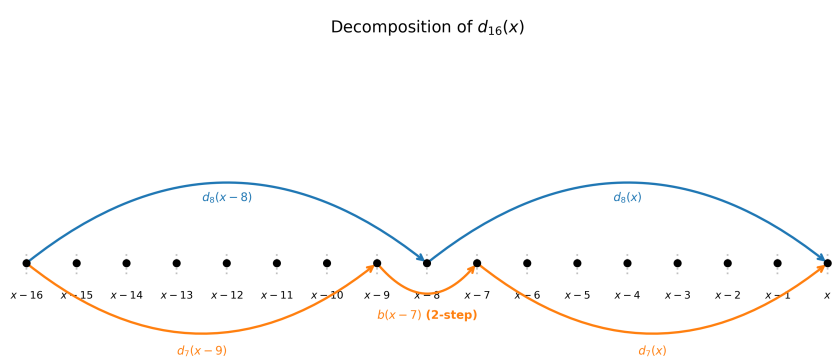

Figure 1: Visualization of the d_{16} decomposition

Building d_k for odd k: inclusion-exclusion

For odd $k = 2m + 1$, we cannot split cleanly at a single midpoint because k is not divisible by 2. Instead, we consider two *overlapping* splits — at $x - m$ and $x - m - 1$ — and correct for double-counting via inclusion-exclusion.

- **Split A** (at $x - m$): paths from $x - k$ to $x - m$, then from $x - m$ to x. Contributes $d_{m+1}(x) \cdot d_m(x - m - 1)$.
- **Split B** (at $x - m - 1$): paths from $x - k$ to $x - m - 1$, then from $x - m - 1$ to x. Contributes $d_m(x) \cdot d_{m+1}(x - m)$.

The two splits overlap on paths that pass through *both* $x - m - 1$ and $x - m$. These are paths that make a 1-step from $x - m - 1$ to $x - m$, with weight $a(x - m)$. Subtracting the overlap:

$$d_{2m+1}(x) = d_{m+1}(x) \cdot d_m(x-m-1) + d_m(x) \cdot d_{m+1}(x-m) - d_m(x) \cdot a(x-m) \cdot d_m(x-m-1)$$

For example, at window size 15 (i.e., $m = 3$):

$$d_{15}(x) = d_8(x) \cdot d_7(x - 8) + d_7(x) \cdot d_8(x - 7) - d_7(x) \cdot a(x - 7) \cdot d_7(x - 8)$$

This three-term formula has a geometric reading: count paths through the left boundary, count paths through the right boundary, subtract those counted by both. The subtracted paths are precisely the ones that 1-step across the boundary, which the recurrence weights by a.

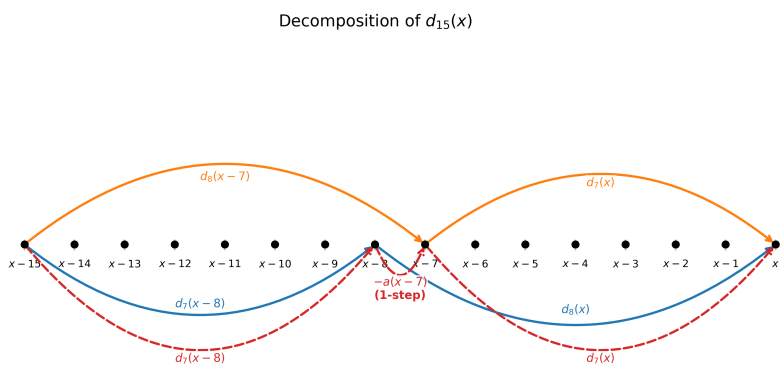

Figure 2: Visualization of the d_{15} decomposition

The Windowed Partial Sums f_k

The arrays $f_k(x)$ play a complementary role. Define $f_k(x)$ as the value f would take at position x if all positions strictly before $x-k$ were set to zero — in other words, the contribution from g-values and from the boundary conditions *within* the local window $[x-k, x]$.

These satisfy a recurrence structurally identical to the d_k formulas. For even window size $2k$:

$$f_{2k}(x) = f_k(x) + d_k(x) \cdot f_k(x-k) + d_{k-1}(x) \cdot b(x-k+1) \cdot f_{k-1}(x-k-1)$$

Reading right to left: $f_k(x)$ captures the right-half contribution; $d_k(x) \cdot f_k(x-k)$ is the left-half contribution propagated forward through paths crossing the midpoint; and $d_{k-1}(x) \cdot b(x-k+1) \cdot f_{k-1}(x-k-1)$ is the left-half contribution propagated through paths that 2-step over the midpoint.

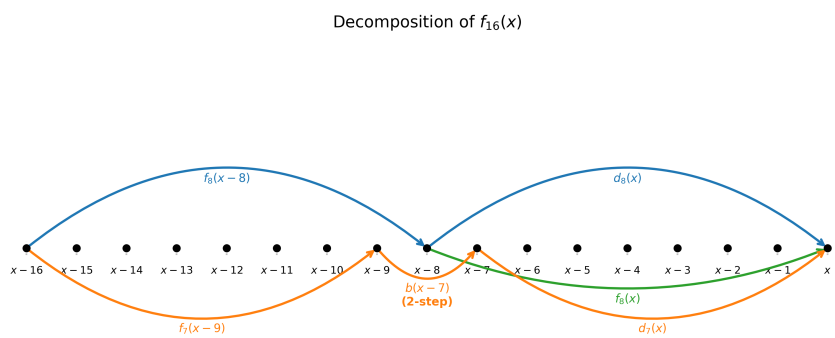

Figure 3: Visualization of the f_{16} decomposition

The odd window size formula is analogous, with a similar inclusion-exclusion structure as d_{2m+1}.

Building the Tower and Stitching Together

The computation proceeds in a logarithmic tower of layers. Starting from the base cases $d_1(x) = a(x)$, $d_0(x) = 1$, $f_1(x) = g(x)$, and $f_0(x) = 0$, each layer doubles the window size, computing d_{2k}, d_{2k-1}, f_{2k}, and f_{2k-1} from d_k, d_{k-1}, f_k, and f_{k-1} of the previous layer. After $\log_2 k$ layers, we have window-16 arrays in hand.

The final pass resolves the full recurrence by stitching windows together:

$$f(x) = f_{16}(x) + d_{16}(x) \cdot f(x-16) + d_{15}(x) \cdot b(x-15) \cdot f(x-17)$$

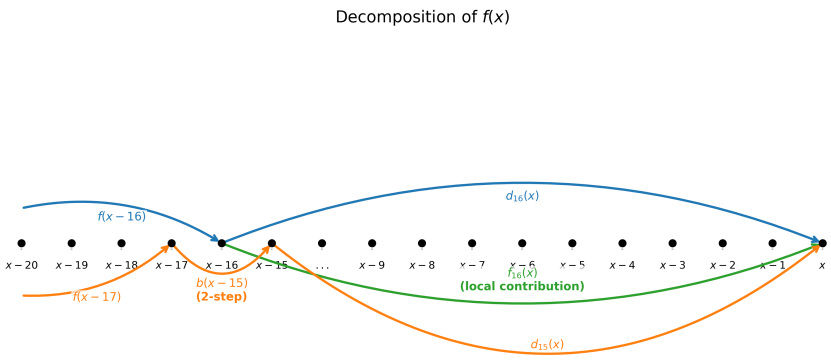

Figure 4: Visualization of the final stitching

The three terms are exactly analogous to the base recurrence itself: $f_{16}(x)$ is the self-contained contribution of the current window, $d_{16}(x) \cdot f(x-16)$ propagates the accumulated history through paths that visit $x-16$, and $d_{15}(x) \cdot b(x-15) \cdot f(x-17)$ propagates history through paths that 2-step over $x-16$ (landing at $x-15$ from $x-17$ with weight $b(x-15)$, then traveling the remaining 15 positions with weight $d_{15}(x)$).

Comparing the construction here with the matrix prefix scan formulation, the coefficient $d_{15}(x) \cdot b(x-15)$ is exactly the $(1,2)$ entry of the combined matrix — the one capturing the indirect influence of $f(x-17)$ via a 2-step over the boundary (otherwise the matrix multiplication form will double count). The local window decomposition recovers this matrix structure implicitly, without ever forming full matrix products.

Work Analysis

At each layer of the tower, we compute exactly 4 arrays (d_{2k}, d_{2k-1}, f_{2k}, f_{2k-1}), each requiring a constant number of operations per element. The tower has $1+\log_2 k$ layers. So the work multiplier compared to scalar evaluation is roughly $\mathbf{4} \cdot (\mathbf{1} + \log \mathbf{k})^2$ per output element.

For $k = 16$, this gives $4 \cdot 5 = 20$, which is better than the matrix scan's $6 \cdot 5 = 30$. The asymptotic depth remains $O(\log n)$ and the total work remains $O(n \log n)$. This suggests this algorithm is still probably not more efficient than the serial

[2]Again this is an approximation because for example $d_1(x) = a(x)$ and we don't need to compute $f_{15}(x)$.

implementation for 32 bits datatypes, but likely worth it for datatypes like `int8` or `int16`.

Extension to Third-Order Recurrences

We also briefly describe how the algorithm extends to higher-order recurrences. For a third-order LTV recurrence

$$f(x) = a(x)\, f(x-1) + b(x)\, f(x-2) + c(x)\, f(x-3) + g(x),$$

the path model extends by adding 3-steps (with weight c). The even-split formula for d_{2k} gains two new terms capturing paths that 3-step over the midpoint:

$$d_{16}(x) = d_8(x)\, d_8(x-8) + d_7(x)\, b(x-7)\, d_7(x-9) + d_7(x)\, c(x-7)\, d_6(x-10) + d_6(x)\, c(x-6)\, d_7(x-9)$$

The first two terms handle the 1-step crossing through $x-8$ and the 2-step crossing over it; the third and fourth terms are new, handling 3-step crossings: one departs from $x-10$ and lands at $x-7$ (weight $c(x-7)$), and the other departs from $x-9$ and lands at $x-6$ (weight $c(x-6)$).

The odd-split formula looks as follows:

$$\begin{aligned} d_{15}(x) = & d_8(x)\, d_7(x-8) + d_7(x)\, d_8(x-7) - d_7(x)\, a(x-7)\, d_7(x-8) \\ & + d_6(x)\, c(x-6)\, d_6(x-7) \end{aligned}$$

The four terms correspond to four cases: paths that pass through $x-7$ but not $x-8$, paths that pass through $x-8$ but not $x-7$, paths that pass through both (the overlap), and paths that pass through neither and instead do 3-step at $x-6$ (the new case).

In the second-order case, a 2-step can skip at most one of the two adjacent positions, so two-term inclusion-exclusion sufficed; with 3-steps available, a single jump can skip both, which correspond to the new case above.

Complexity-wise, each layer of the tower now requires 3 d-arrays (d_k, d_{k-1}, d_{k-2}) and 3 f-arrays — 6 arrays total per layer, versus 4 for the second-order case.

The general pattern seems to be $2p$ arrays per layer for a p-th order recurrence. This matches the intuition: a p-th order recurrence allows p ways to cross any boundary (1-step, 2-step, ..., p-step), and each introduces an independent $d_{k-1}, d_{k-2}, \ldots$ interaction. The inclusion-exclusion correction for odd windows generalizes correspondingly, though the precise formula for $p > 2$ is left as future work.

Compared to the matrix prefix scan, where a p-th order recurrence requires a $(p+1) \times (p+1)$ matrix with $p(p+1)$ nontrivial entries, the local window decomposition's $2p$ arrays per layer scales much more favorably. For $p = 2$, the matrix approach needs 6 entries versus our 4 arrays; for $p = 3$, the comparison is 12 versus 6.

Closing Thoughts

This algorithm replaces the sequential dependency chain with a parallel tower-building phase while tracking fewer arrays per layer than a matrix prefix scan. One helpful intuition for this improvement is that the matrix form of the recurrence is sparse, and the six non-constant entries compute overlapping information.

As for future work, I have not formalized the general recipe for order-p recurrences and, on the experimental side, I have not measured the actual performance of this improvement and the original matrix scan and understood the trade-offs in practice.